

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Performed **physical, chemical and chromatographic testing** on enterprise laboratory systems using **SQL** and **Python** data pipelines to support product release, raw material analysis, and compendia testing workflows; maintained **cGMP and ISO compliance** across all analytical procedures and achieved **100% audit readiness** through systematic documentation controls.
- Prepared and updated **Control forms, SOIs, and lab documentation** with **Power BI** dashboards and **DAX** measures to track testing trends and generate **technical reports** from analytical data; supported lab management in trend analysis and regulatory submissions with **95% accuracy** in data validation.
- Participated in **lab projects including Method Transfers, Verifications, Validations and Equipment Qualifications** using **change-control management** and **cross-functional collaboration** with laboratory teams; coordinated **UAT** and post-implementation support to ensure all analytical instruments met **ISO compliance** and qualification standards.
- Identified, investigated and resolved **out-of-specification (OOS)** problems in chemistry lab operations using **root-cause analysis** and statistical methods; prepared detailed investigation reports with corrective actions, reducing lab turnaround time by **20%** and maintaining continuous improvement in analytical processes.
- Supported **audit preparations** and maintained an **audit-ready lab environment** through rigorous adherence to **cGMP regulations, Environmental, Health and Safety regulations**, and the company's **Guiding Principles and Code of Conduct**; documented all laboratory activities and maintained **ISO compliance** records for regulatory inspections.
- Developed competence in **analytical techniques** and laboratory operations through hands-on testing of raw materials, finished goods, and maintenance protocols; demonstrated ability to work as a dedicated team member to ensure laboratory functions operated at peak efficiency and met all **cGMP and ISO compliance** requirements.

SKILLS

Analytical Testing & Chemistry Lab – physical testing, chemical testing, chromatographic testing, raw material analysis, finished goods testing, compendia testing, analytical techniques.

Laboratory Compliance & Quality – cGMP, ISO compliance, audit preparation, audit-ready environments, Environmental Health and Safety regulations, Code of Conduct.

Lab Documentation & Reporting – Control forms, SOIs, lab documentation, technical reports, trend analysis, data validation, regulatory submissions.

Laboratory Projects & Change Management – Method Transfers, Verifications, Validations, Equipment Qualifications, change-control management, UAT, post-implementation support.

Data Analysis & BI Tools – Power BI, DAX, SQL, Python, data modelling, statistical analysis, root-cause analysis, trend visualization.

Problem-Solving & Compliance – out-of-specification (OOS) investigation, corrective actions, continuous improvement, laboratory team collaboration, regulatory compliance.

EDUCATION AND TRAINING

PROJECTS

Pharmaceutical Chemistry Lab Testing & Compliance Automation Platform (HCL Technologies)

- Designed and deployed an integrated **laboratory testing workflow automation system** across raw material, finished goods, and equipment maintenance operations, leveraging **Python ETL pipelines** to collect analytical test data, validate against compendia specifications, and flag **out-of-specification (OOS) results** in real-time; reduced manual testing documentation time by **40%**.
- Built **Power BI dashboards** with **DAX measures** and **SQL Server** backend to visualize chromatographic results, physical property trends, and chemical composition metrics across **1,200+ monthly tests**; automated **technical report generation** for lab management and regulatory submissions with **99.2% accuracy**.
- Implemented change-control and audit-trail logging using **SQL stored procedures** to track all Control forms, SOIs, and method verification updates; ensured **cGMP and ISO compliance** across all lab documentation, achieving **zero audit findings** in external regulatory inspections.
- Developed **root-cause analysis automation** using **Python statistical methods** (hypothesis testing, regression analysis) to investigate **OOS incidents**; generated investigative reports with corrective-action recommendations, reducing lab turnaround time from **14 to 11 days (21% improvement)** and supporting continuous process improvement.
- Coordinated **Method Transfers, Validations, and Equipment Qualifications** across lab infrastructure; maintained **audit-ready environment** documentation and **Environmental, Health and Safety compliance** logs, supporting seamless third-party audits and regulatory inspections with **zero non-conformances**.

Lab Incident Investigation & Compliance Reporting Dashboard (HCL Technologies)

- Created a **Tableau-based compliance dashboard** to track and visualize **out-of-specification (OOS) investigations**, corrective actions, and lab incident resolution timelines; enabled lab management to identify systemic analytical issues and prioritize process improvements.
- Automated **cGMP audit preparation** workflows using **Python scripting** to aggregate Control forms, SOIs, method qualification records, and equipment maintenance logs into a single audit-ready repository; reduced audit preparation time by **35%** and maintained **ISO compliance** documentation across all laboratory functions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)